
Welcome
To Our
Presentation

Title:

Understanding Customer Purchase Pattern in a Coffee Shop



Presentation by: Girija Shivanand Desai

Course: Bachelor of Computer Applications

Academic Year: 2026

College Name: S.M.G.K.M College, Sangli



www.reallygreatsite.com

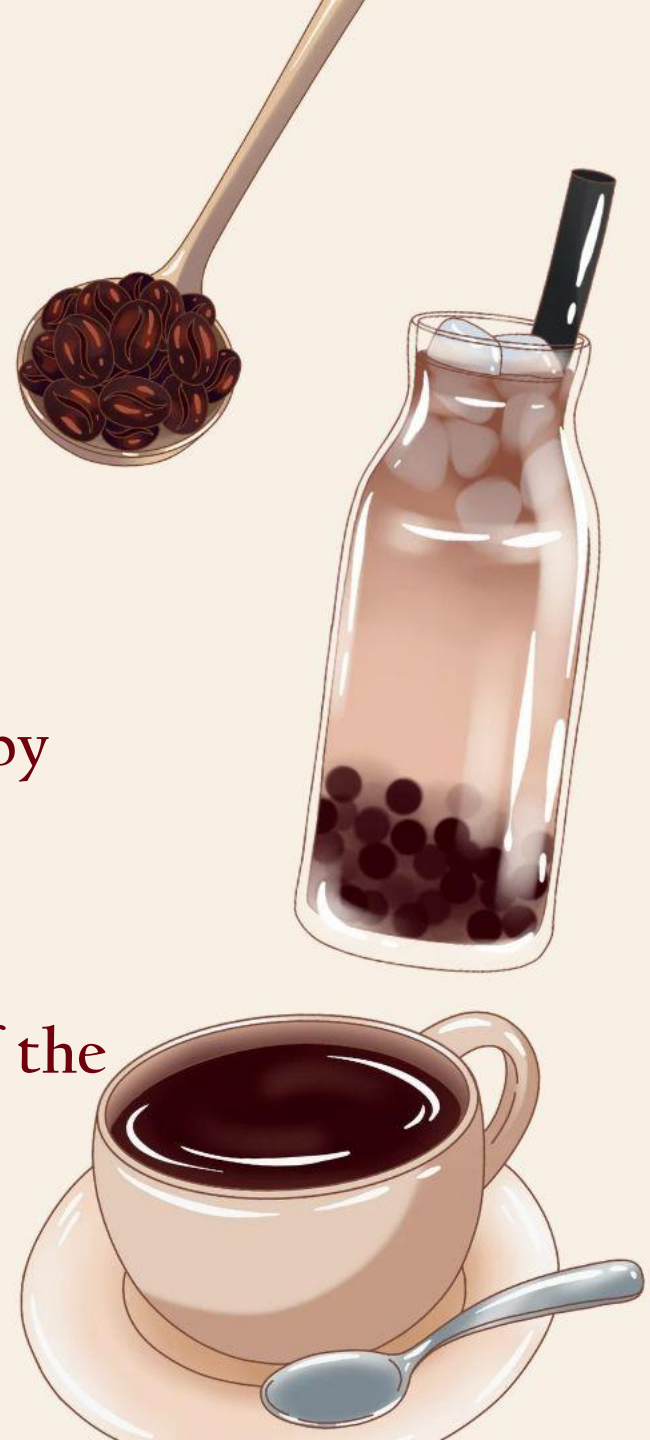
Data Collection:

Data collection contains transaction level records, including order details, product information, quantities sold, price, store locations, and purchase dates for customer purchase pattern analysis.

Python
Import pandas as pd
Import numpy as np

df=pd.read_excel("Coffee_Shop_Sales.xlsx")
df=head()

- Import pandas and numpy library.
- Loads the CSV file into a DataFrame.
- Shows the first 5 rows of the dataset.



Objectives:

- Identify peak customer purchasing times.
- Evaluate product performance.
- Analyze customer purchase behaviour.
- Compare store location performance.
- Revenue and profitability analysis.
- Analyze overall sales performance.



Data Description:

- Includes both product-level and time-based information.
- Suitable for Exploratory Data Analysis(EDA),Dashboard development, forecasting and customer behavior studies.
- Unique identifiers for each transaction.
- Unit number within the store.
- Contains approximately 149,000+ transaction records.
- Represents sales across multiple store locations.



Data Cleaning Steps:

1. Load the Dataset
2. Check missing values.
3. Remove duplicates records.
4. Standardize column names.
5. Fix data types.
6. Check invalid values.
7. A Remove extra.
8. Standardize text format.
9. Check outliers.
10. A Create revenue column.
11. Extra date features.
12. Check unique categories.
13. Final validation.



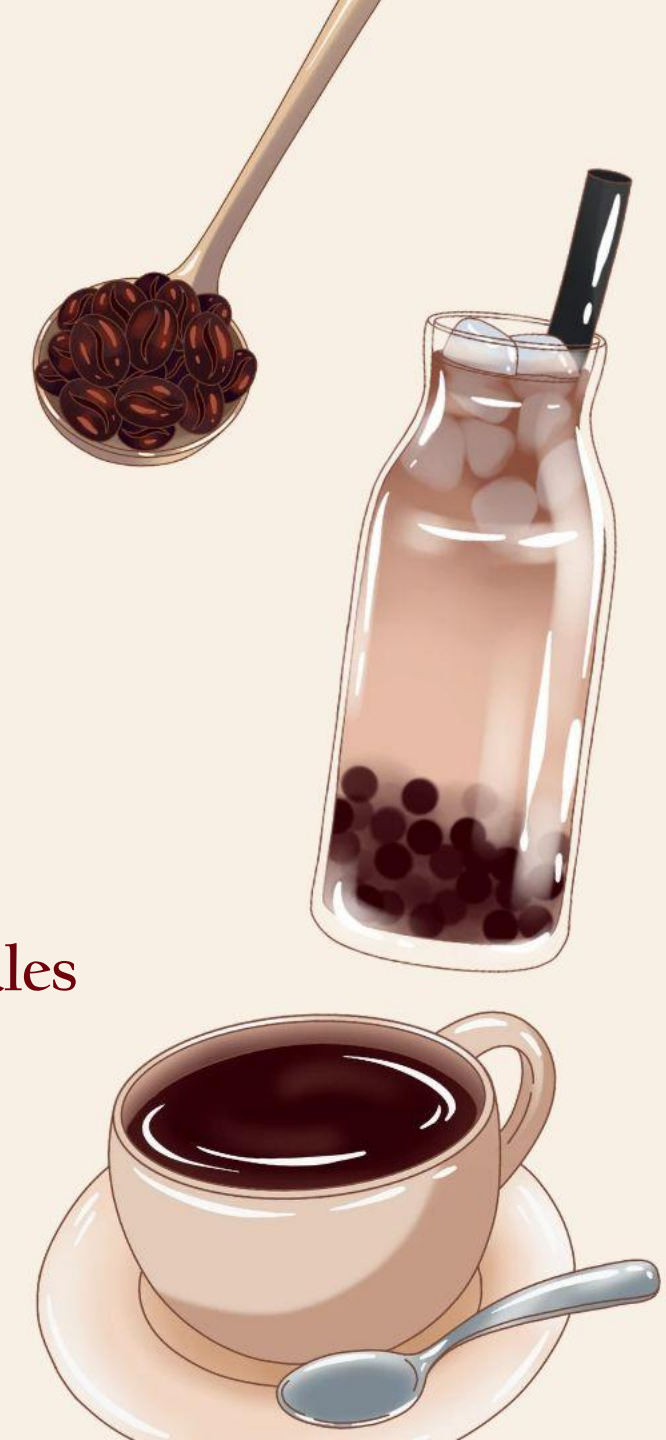
Data Preprocessing:

1. Imported the dataset using Pandas.
2. Checked the dataset shape, data types, and summary statistics.
3. Identified and removed duplicate records.
4. Checked for missing values and handled them if present.
5. Converted transaction_time to time format.
6. Converted transaction_time to time format.
7. Corrected inconsistent data types for numerical and categorical



Feature Engineering:

- Created Sales=Transaction qty * unit price.
- Extracted Year, Month, Day, and Day of Week from the transaction date.
- Extracted hour from the transaction time.
- Created a Time of Day feature(Morning, Afternoon, Evening).
- Calculated total sales by Store Location, Product Category and Product Type for analysis.
- Created revenue-based features such as average sales per transaction and
- Daily sales totals.



Feature Scaling:

- Selected numerical features such as:

1. transaction_qty
2. Unit_price
3. Sales

- Applied StandardScaler(Z-score normalization) to standardize numerical features for machine learning.

- Alternatively, MinMaxScaler can be used to scale values between 0 to 1 when requried by certain algorithm
- Categorical features such as Store Location, Product Category, and Product Type can be encoded using Label Encoding or One-Hot Encoding before model training.



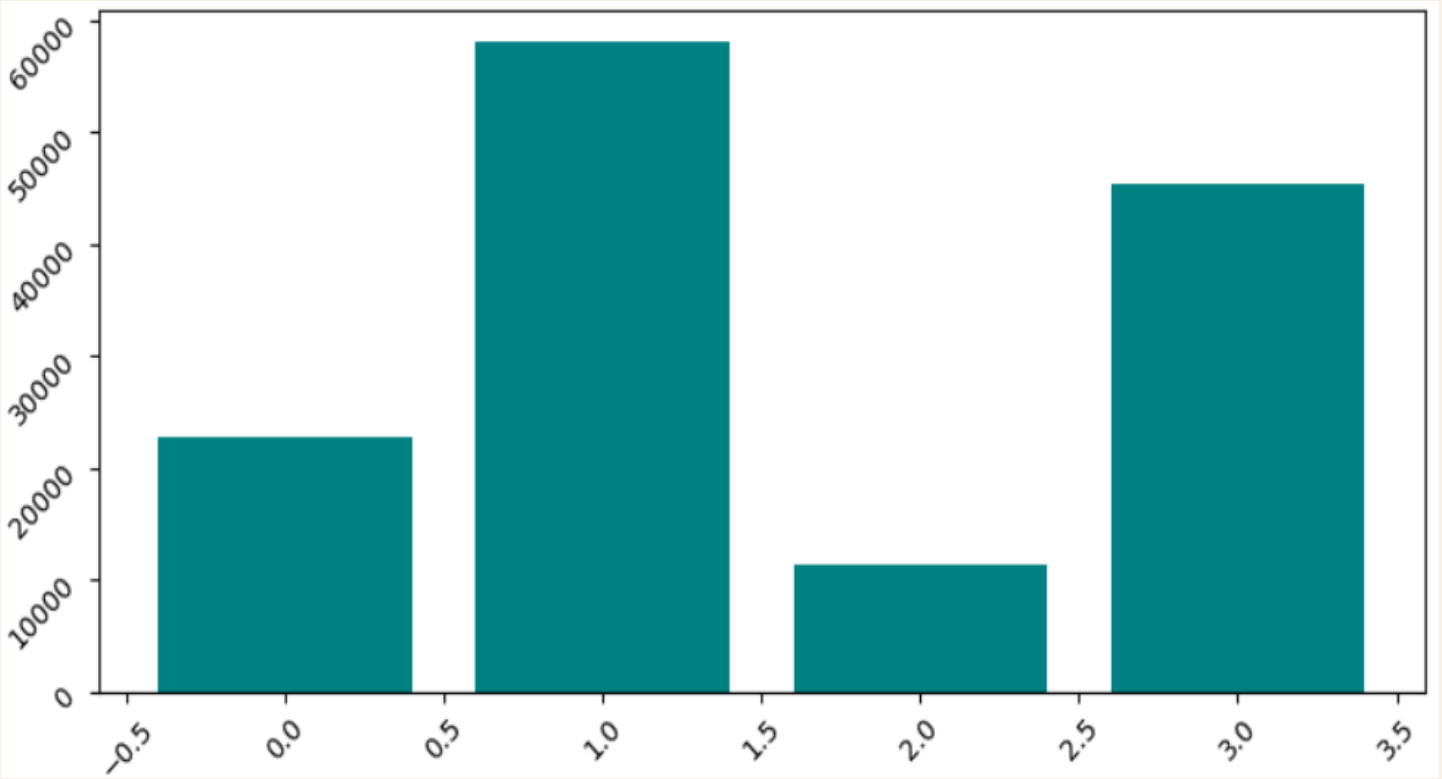
Visualization:

- Visualization is a process of presenting data in the form of charts and graph to identify patterns ,trends and insights easily.
- Easy understanding: Makes complex data simple to understand.
- Identifies patterns: Helps find patterns and relationships in data.
- Shows trends: Displays changes in data over time.
- Supports decision-making: Helps people make better decisions.
- Saves time: Presents large amounts of information quickly.
- Improves communication: Makes information easier to explain and present.



a)Revenue by Product Quality:

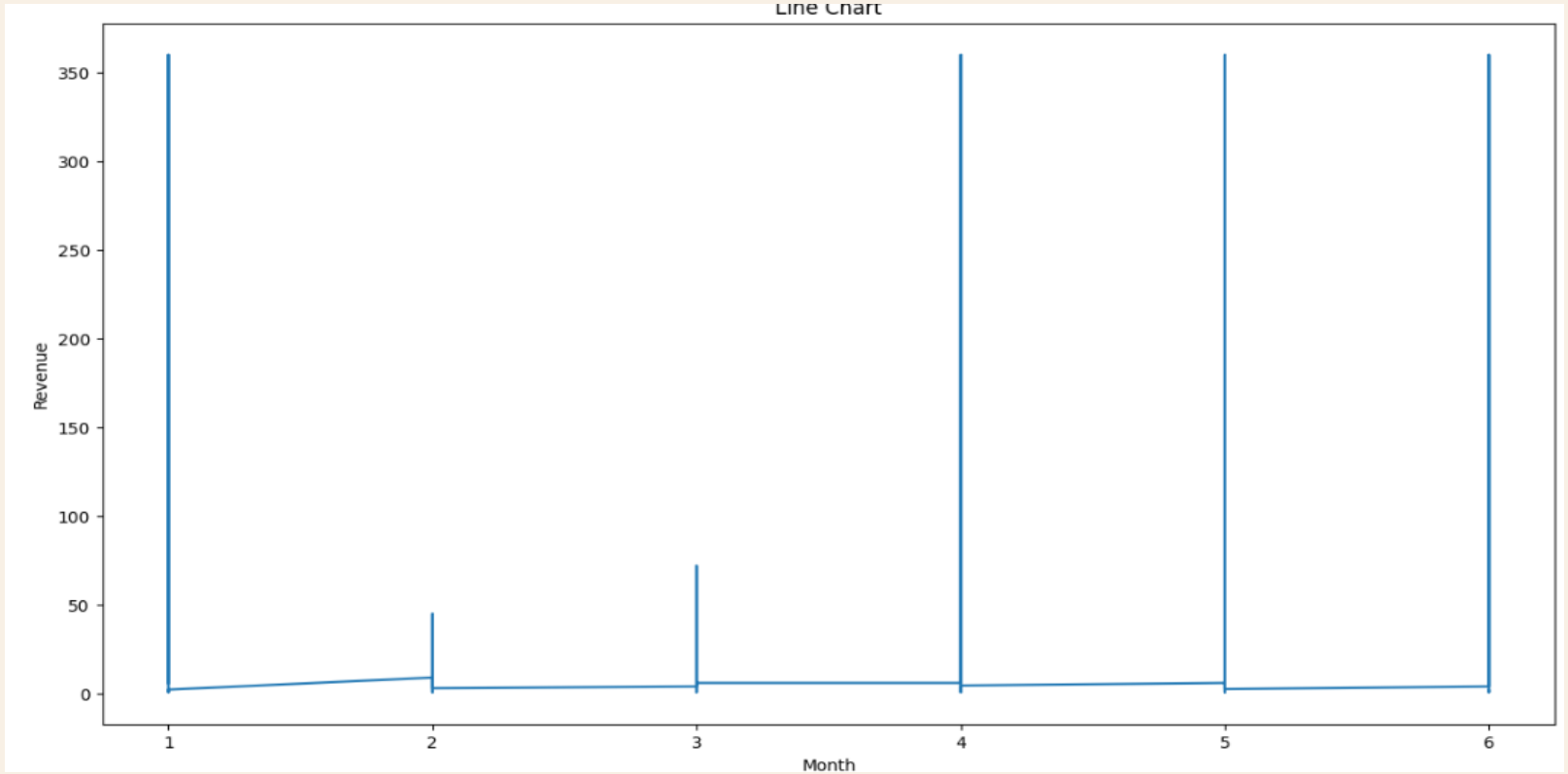
This bar chart shows the relationship between product quantity and revenue. The X-axis represents product quantity, and the Y-axis represents the revenue generated.





b)Monthly Revenue Analysis:

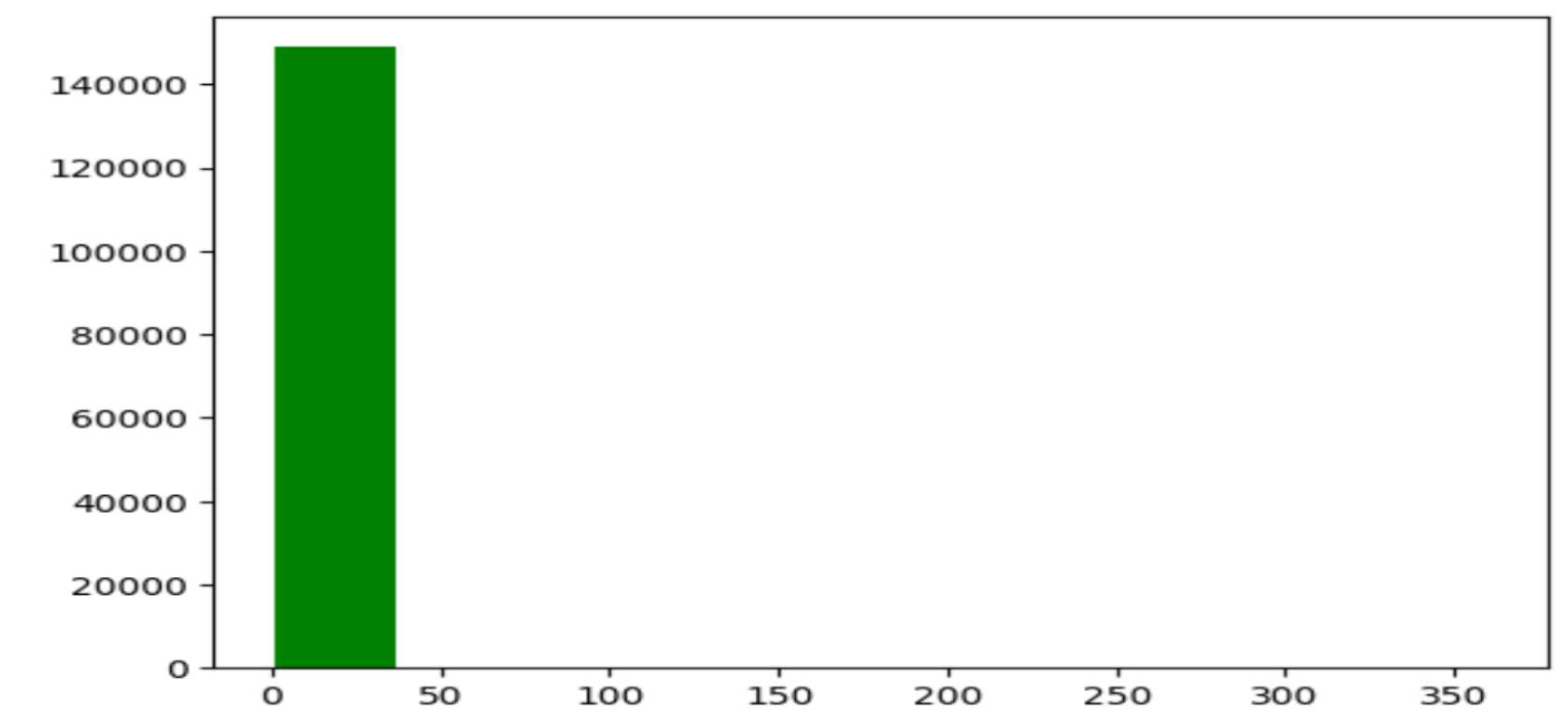
The chart helps businesses understand monthly revenue trends, identify their best-performing months, and make informed decisions to improve overall sales performance.





c)Revenue Frequency Distribution:

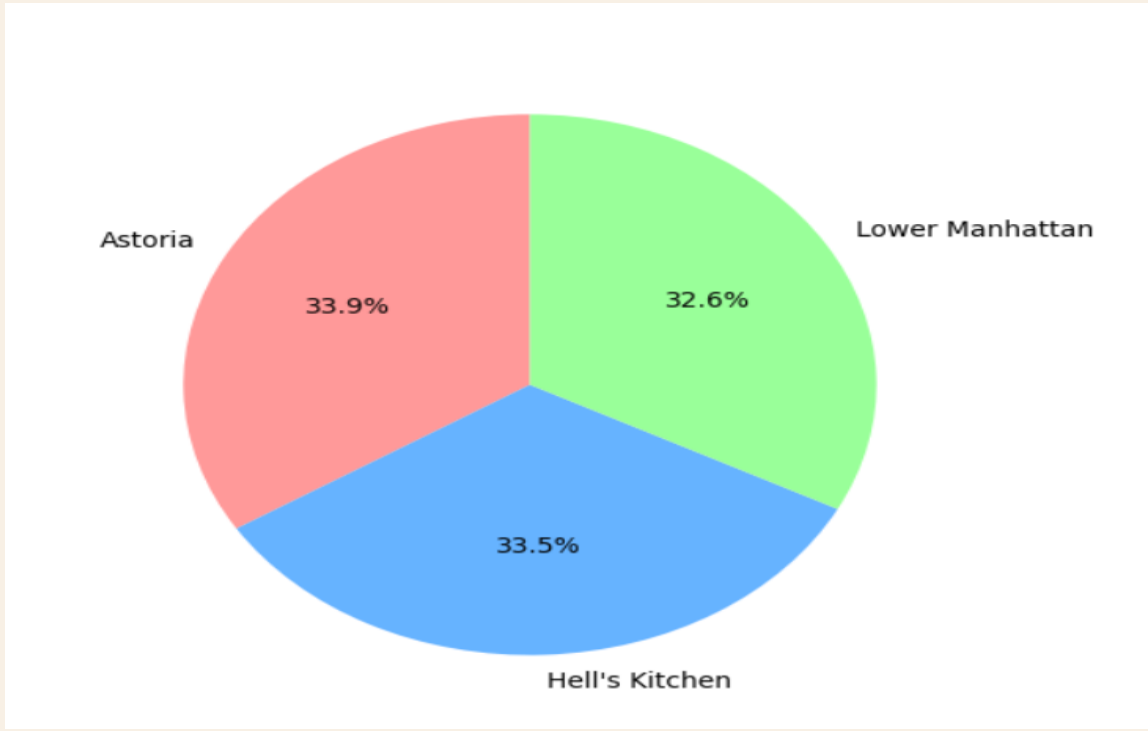
This histogram shows the distribution of revenue values and how frequently they occur. It helps identify the most common revenue range and understand the overall pattern of revenue in the dataset.





d)Total Sales Revenue by Store Location:

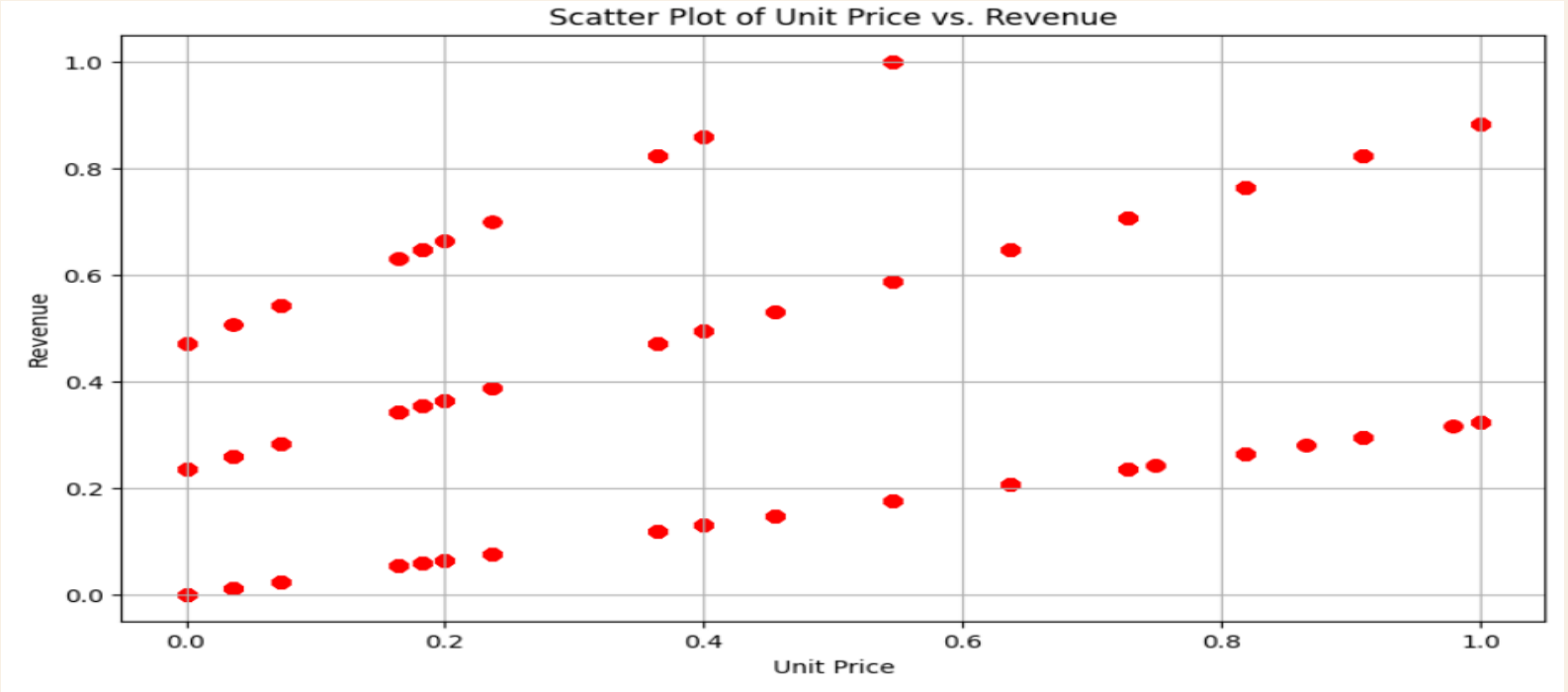
This pie chart represents the contribution of each store location to the total sales revenue. It helps compare store performance, identify the highest revenue-generating location, and understand each store's share of overall sales.





e) Unit Price by Revenue:

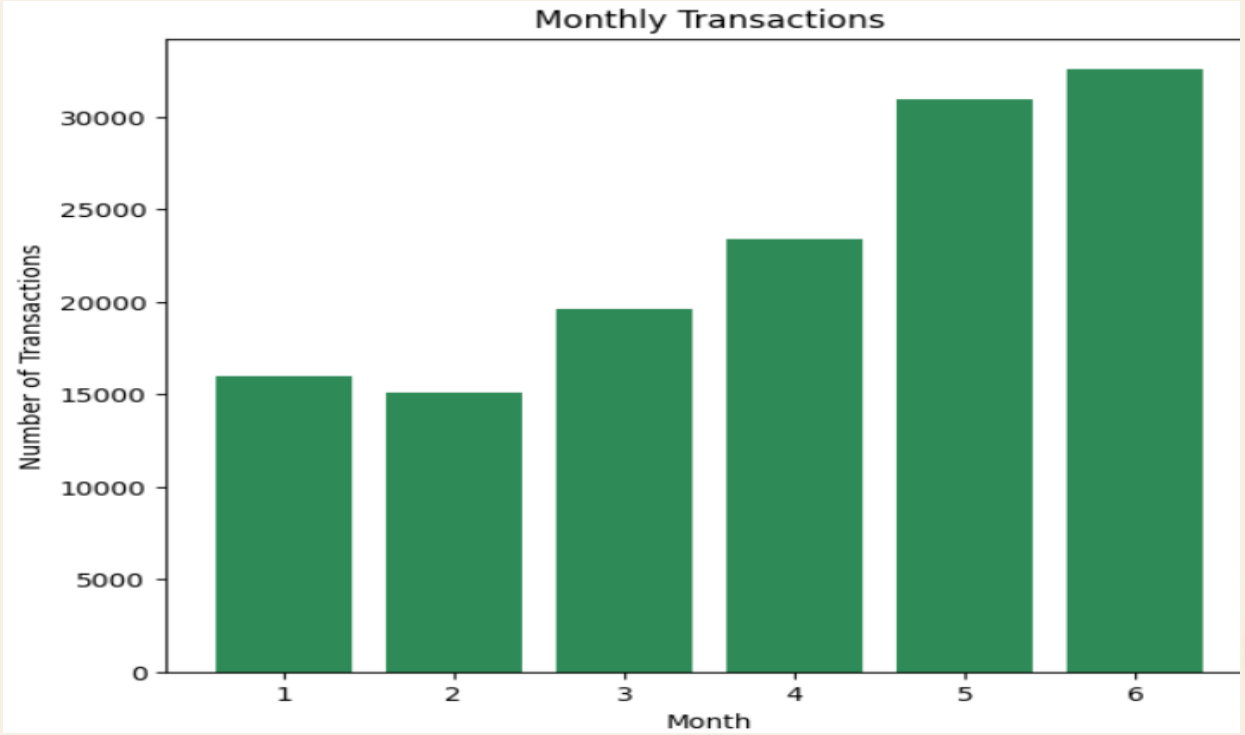
This scatter plot shows the relationship between unit price and revenue. It helps identify whether products with higher unit prices tend to generate higher revenue, highlights patterns in sales performance, and identifies unusual data points.





f) Monthly Transaction Analysis:

This bar chart shows the number of transactions recorded each month. It helps compare monthly transaction activity, identify the busiest and least active months, and understand trends in customer purchasing behavior.





Insights:

- Coffee products contribute the largest share of total sales revenue.
- Sales peak during morning hours, indicating strong breakfast-time demand.
- Weekdays generally generate higher transaction volumes than weekends.
- A small group of top-selling products accounts for a significant portion of revenue.
- Store performance varies by location, highlighting differences in customer traffic.
- Most customers purchase only one or two items per transaction.
- Product categories such as coffee and tea dominate overall sales volume.



Conclusion:

- Customer purchase behavior is strongly influenced by time of day, product category,store location and seasonality.

- By analyzing these patterns, coffee shops can improve operational efficiency, enhance customer satisfaction , and maximize revenue through informed business decisions.

- Coffee-based beverages are the most popular products and Sales are highest during peak morning and afternoon hours.



Thank You













